Phantom Q Security Lab

Attack, defend, verify and learn secure communication

An interactive cybersecurity learning platform where learners experience how secure communication is protected, attacked, verified and improved.

CORE CONCEPTS

- Foundations
- Symmetric Security
- Asymmetric Security
- Quantum Key Distribution

Built for **Missions, Competition, Practice/Lab** and **Classroom** use.

The Core Problem We Are Solving

From theory-only learning to practical, role-based cybersecurity understanding

Traditional learning challenge

Learners often read about cybersecurity, but do not experience what happens when information is protected, intercepted, attacked or verified.

- Theory is remembered, but not always understood in action
- Roles such as sender, receiver and attacker remain abstract
- Awareness of suspicious behaviour is difficult to build from notes alone

Phantom Q approach

Phantom Q turns secure communication into an interactive experience where learners protect, attack, verify and respond.

- See security events happen in context
- Learn through attack–defence interaction
- Connect simulation, awareness and engineering practice

Theory only → limited understanding | **Interactive role-based security →** practical awareness and stronger learning

The Four Core Security Concepts

The educational backbone of the Phantom Q platform

These four concepts form the progression path for **Missions, Practice/Lab** and **advanced security challenges.**

1 Foundations

Introduces the basic idea that information can be transformed and protected.

Learning Focus
Text shifts, image scrambling and protection awareness

2 Symmetric Security

Explains how Alice and Bob use one shared secret key.

Learning Focus
Shared-key protection, transmission and interception risk

3 Asymmetric Security

Shows how public and private keys change the way security is established.

Learning Focus
Public lock, private access and mathematical attack pressure

4 Quantum Key Distribution

Introduces quantum key establishment and trust through detectable disturbance.

Learning Focus
QKD simulation, hardware link and trusted-key decisions

1010 0101 **Information transformation** → **Shared secret** → **Public/private security** → **Quantum key establishment**

Attack and Defence Are Both Part of Learning

Phantom Q is not one-sided: learners build security by understanding both protection and attack

Defence Side

Alice

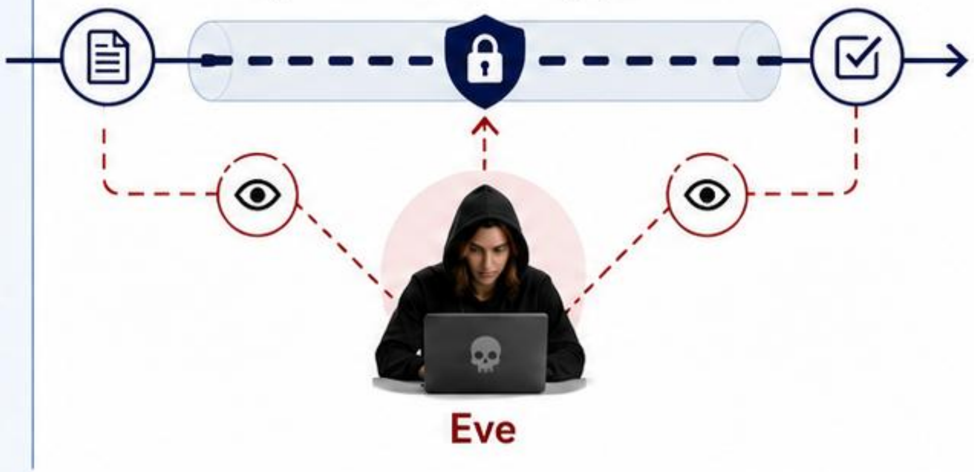
Prepare / Protect / Send

Bob

Receive / Verify / Recover

Protected Communication Path

Encrypted • Authenticated • Integrity Protected



Attack Side

Eve

Observe / Intercept / Analyse / Exploit

To **defend** properly, learners must understand how attacks work.
To **attack responsibly**, learners must understand what defenders can **detect**.

Role progression
Roles are strict during a live activity, but learners may rotate across Alice, Bob and Eve over time.

Player knowledge may broaden across activities; live information remains role-specific.

Phantom Q supports both specialist depth **and** generalist breadth in cybersecurity.

- Defence (Blue)
- Attack (Red)

Both perspectives, one mission.

Role Families: Alice, Bob and Eve

Three core security perspectives inside Phantom Q

Alice

Protection / Sender

Main function

Prepares and sends protected information

Core actions

- Prepare
- Protect
- Send

Information note

Knows the original message or state being prepared

Bob

Verification / Receiver

Main function

Receives, checks and recovers information

Core actions

- Receive
- Verify
- Recover

Information note

Knows the received outcome and verification evidence

Eve

Adversarial / Attacker

Main function

Observes, intercepts and tests weaknesses

Core actions

- Observe
- Intercept
- Analyse
- Exploit

Information note

Does not automatically know Alice's secret preparation

Roles are strict during a live activity, but learners may rotate across Alice, Bob and Eve over time.

Player knowledge may broaden across activities; live information remains role-specific.

- Alice and Bob form the defence/protection side
- Eve represents the adversarial/testing side
- All three roles are essential to cybersecurity learning

Flexible Player Configurations

Phantom Q is built around security roles, not fixed player count

Security roles are persistent. Human occupancy is flexible.

Individual Play
Learn at your own pace

1. **Solo** — One learner with missing roles controlled by the system
2. **1v1** (vs) — Defender vs attacker
3. **2v1** (vs) — Two learners on one side, one on the other

Small Group Play
Share roles, learn together

4. **Tri-role** — Alice, Bob and Eve each played by one learner
5. **3 Defenders vs 3 Attackers** (vs) — Dedicated Red-vs-Blue team format

Team / Classroom Play
Scale learning across teams and classrooms

6. **Trio vs Trio** (vs) — Alice + Bob + Eve versus Alice + Bob + Eve
7. **Classroom Groups** — Multiple teams or stations operating in parallel

Roles Can Be Filled By

Human
A learner actively plays a security role.

System-controlled role
The system fills the role when no human is available.

Team
Two or more learners share and coordinate a security role.

Roles persist in the environment regardless of who or what occupies them.

- The platform does not require exactly three human players
- Missing roles can be system-controlled
- The same security concepts scale from solo play to classroom play

Experience Modes

How learners engage with Phantom Q across structured learning, play and teaching.

1 Missions

Structured progression through the security concepts

- Learning objectives and guided activities
- Supports solo or cooperative play
- Leads naturally into storyline and campaign structure

2 Competition

Player and team-based attack–defence challenges

- Supports 1v1, team and tri-role formats
- Compares security performance across sides
- Used for challenge, practice and replayability

3 Practice / Lab

Experiment with concepts, skills and hardware without mission pressure

- Try symmetric, asymmetric and QKD systems directly
- Useful for testing, training and demonstrations
- Supports physical hardware interaction

4 Classroom / Host

Teacher-led setup, control and group coordination

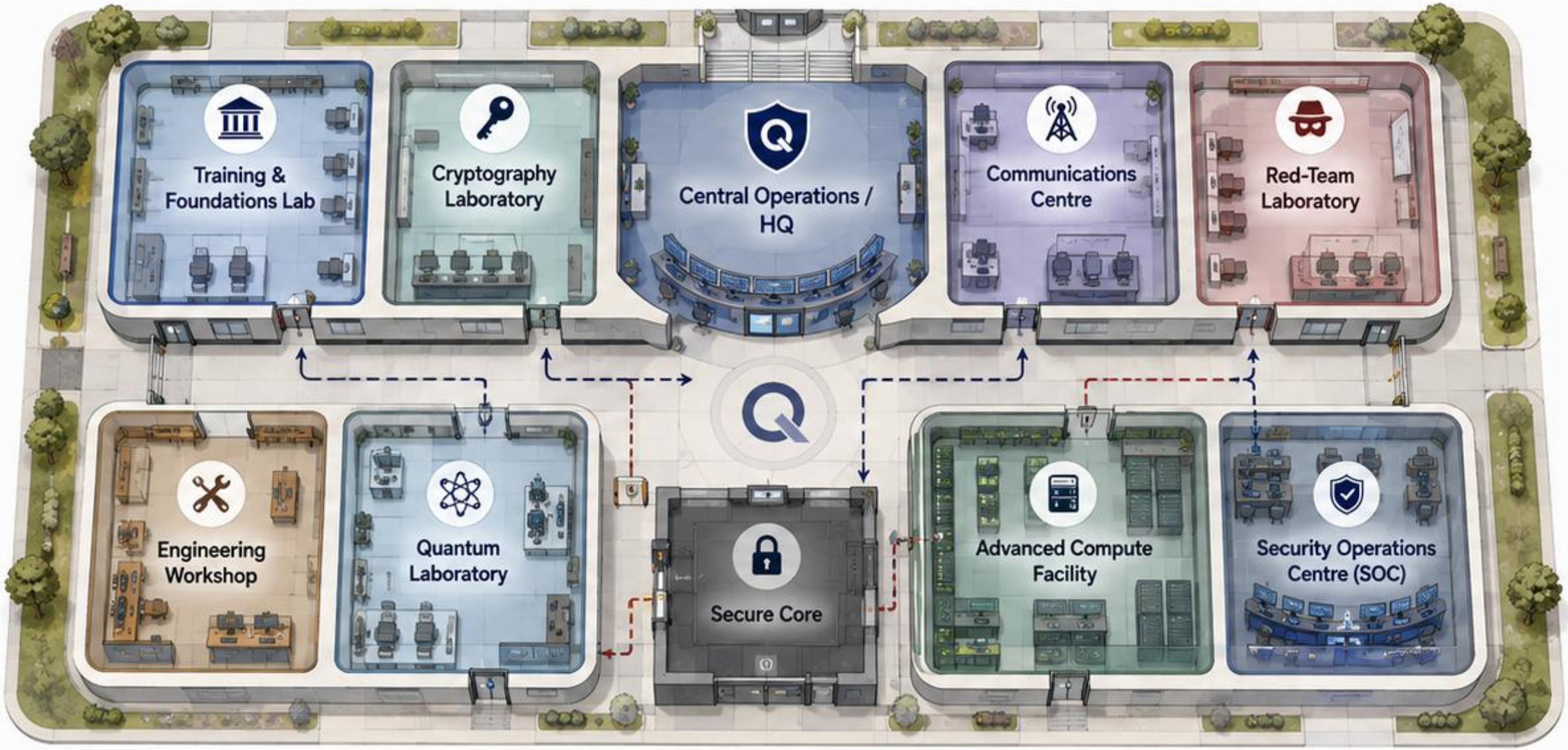
- Create and manage sessions
- Assign or rotate roles
- Observe groups and support guided learning

Multiplayer is a capability, not a separate experience mode.
Missions, Competition and Practice/Lab can all support single-player or multi-user participation depending on the activity.

These modes separate learning, competition, experimentation and teaching without overcrowding the user experience.

Phantom Q Headquarters

A shared security-learning establishment where rooms represent real capabilities



HEADQUARTERS PRINCIPLES

- One shared headquarters
- Different rooms represent different capabilities
- Progression unlocks access, not just decoration
- Location gives context to activity

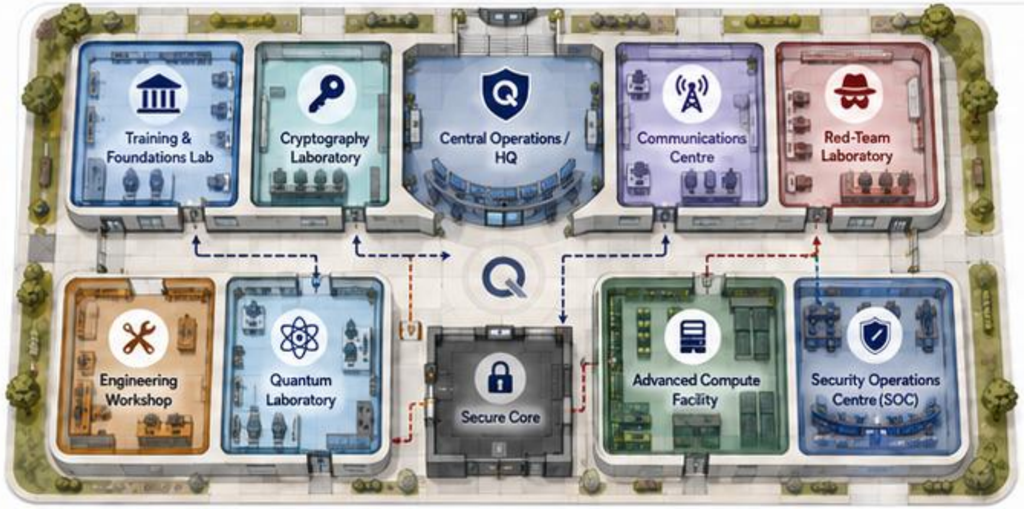
ACCESS LEVELS

- Standard access
- Restricted access
- High-security / endgame access

The Headquarters becomes both the navigation foundation and part of the cybersecurrity experience itself.
Where a player was, what they accessed and whether they were authorised can later become evidence.

Headquarters Capability Map

How each Headquarters facility contributes to the Phantom Q platform



CAPABILITY CATEGORIES

LEARN	Build knowledge and core security understanding.
OPERATE	Use secure communications and platform tools.
ATTACK / TEST	Adversarial testing, attacks and offensive skills.
MONITOR / RESPOND	Detect, investigate and respond to incidents.
ENGINEER	Design, build and connect physical and digital systems.

1 Training & Foundations Lab
- Foundations
- Basic assessment
- Introductory security concepts
- Future prior-knowledge checks

2 Cryptography Laboratory
- Symmetric and asymmetric security
- Keys
- Text protection
- Image protection
- Classical cryptographic experimentation

3 Communications Centre
- Real text transfer
- Image transfer
- Communication channels
- Sender / receiver operation
- Delivery and secure transfer testing

4 Red-Team Laboratory
- Authorised adversarial testing
- Eve capabilities
- Vulnerability testing
- Interception
- Attack tools and offensive skill development

5 Security Operations Centre (SOC)
- Monitoring
- Alerts
- Evidence
- Investigation
- Defensive awareness and incident response

6 Engineering Workshop
- Hardware
- Electronics
- Devices
- Embedded systems
- Bridge between software and physical engineering

7 Quantum Laboratory
- QKD
- Simulation
- Physical Alice/Bob demonstrator
- Quantum-channel experimentation
- Trusted key establishment

8 Advanced Compute Facility
- High-performance compute
- Quantum-computer threat demonstration
- Advanced computational attack capability

9 Secure Core
- Highest-value protected systems
- Endgame missions
- Restricted access
- Final Phantom target

Headquarters rooms represent real platform capabilities, not decorative spaces.

Each facility delivers a distinct product purpose that underpins learning, gameplay and evidence-based incident understanding.

Progression: Skill -> Clearance -> Access

How player growth unlocks responsibility, tools and deeper Headquarters access

1. Skill Demonstrated
Learn concepts, complete activities and demonstrate understanding.

→

2. Capability Proven
Apply knowledge in practice, solve problems and deliver reliable results.

→

3. Security Clearance
Earn trust through consistent performance and responsible behaviour.

→

4. New Rooms / Tools / Missions Unlock
Gain access to higher areas, powerful tools and more complex missions.

Clearance & Advancement Tiers (Examples)

TIER	MEANING	WHAT IT UNLOCKS (EXAMPLES)
Trainee	Entering the program and learning the foundations.	Basic training and introductory access to core facilities (Training Lab, Engineering Workshop).
Operator	Trusted with daily operations and routine tasks.	Communications Centre access and routine secure tasks. Can participate in standard missions.
Security Analyst	Demonstrates analytical ability and investigative skill.	Monitoring systems, evidence suite and investigation tools. Access to threat intel and case activity.
Specialist	Expert in specific disciplines and advanced techniques.	Advanced attack/defence systmes and deeper cryptographic work. Access to specialist labs.
Quantum Security Engineer	Mastery of quantum secure systems and operations.	Quantum Laboratory access and QKD operations. Work with quantum communications and research systems.
Phantom Response	Highest trust and incident command authority.	Highest incident authority. Secure Core access and endgame responsibility. Leads critical response.

Equipmentt & Expertise

Equipment upgrades unlock capability but do not substitute for expertise.

Workstation → **Analysis Workstation** → **Advanced Compute** → **Quantum Resource** → **QKD System**

Workstation	Analysis Workstation	Advanced Compute	Quantum Resource	QKD System
Basic tools for learning and practice.	Stronger tools for analysis and investigation.	High-performance systems for complex tasks.	Quantum-enabled capability for specialist work.	Quantum key distribution infrastructure.

Better tools extend what you can do — expertise determines what you should do.
Skill, judgement and responsibility remain irreplaceable.

Looking Ahead (Future Consideration)

Advanced learners who already know more can later prove outside knowledge through optional challenge or assessment paths, instead of being artificially capped by the base progression.

Planned feature — details to be confirmed at a later time.

Progression is earned through demonstrated skill and trusted responsbbility — not wealth, status or decoration.

Learn → Practice → Prove → Earn Trust → Gain Access → Protect Together

The Phantom: Endgame Adversary

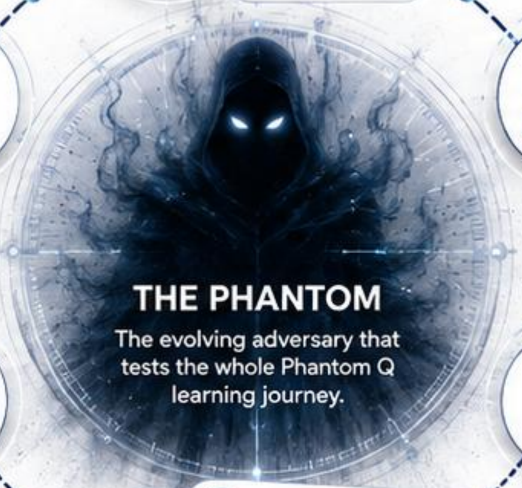
The final boss concept that tests the full security stack

Why not just Eve?

Eve remains the attacker role learners may play.

The Phantom sits above normal role play as the ultimate, adaptive adversary.

The endgame must work whether the learner has played Alice, Bob or Eve.



1 Awareness / identity / suspicious behaviour

Detect deception, phishing, social engineering and insider risk.

2 Symmetric weakness / shared-secret exposure

Exploit weak ciphers, poor key management and secret leakage pathways.

3 Asymmetric weakness / public-key challenge

Target key exchange flaws, weak validation and certificate vulnerabilities.

4 Quantum-computer threat

Leverage quantum advantage to break, accelerate or undermine cryptography.

5 QKD trust and verification challenge

Subvert device trust, tamper with channels and challenge QKD assumptions.

6 Incident response and containment

Create time pressure, force triage and test coordination under real-world stress.

Endgame Perspectives

Defender perspective

Contain or defeat the current Phantom incident.

Attacker perspective

Attempt to penetrate the hardened Phantom Q environment.

Evolving Phantom

A specific Phantom incident can be defeated to complete a campaign, but the wider threat persists and evolves for future challenges and releases.

The endgame challenge unifies awareness, classical cryptography, QKD and incident response into one coherent final test.

Defend the stack. Prove the skills. Earn the victory.

Why QKD Becomes the Climax

From quantum-computer threat to a new key-establishment model

1 Classical public-key / asymmetric security
Security rests on hard mathematical problems.

→

2 Quantum computer breaks the maths
A quantum computer can solve the hard problem and recover the private key.

→

3 QKD changes the problem
Key establishment is based on quantum physics, not the same computational assumption.

→

4 Attacker must shift tactics
The same computational attack does not reveal the QKD key.

Model Comparison

Asymmetric under quantum threat

Public / Private key model
Security relies on asymmetric cryptography.

Hard mathematical problem
$N = p \times q$
Relies on factorisation, discrete log, or similar computational hardness.

Quantum computer attacks the maths
An adversary with a large quantum computer can solve the problem and recover the key.

VS

QKD

Quantum-state-based key establishment
Security comes from the laws of quantum mechanics, not computational difficulty.

No direct factorisation-style target
There is no hard mathematical problem whose solution yields the key.

Trust depends on disturbance checks and verification
Alice and Bob detect eavesdropping through error rates and only keep verified keys.

Attack Surface Shift (Post-QKD)

1. **Channel interference / intercept–resend**
Eve injects signals or performs intercept-resend, causing detectable disturbance and key rejection.

2. **Device attacks**
Side channels, Trojan horse, or physical attacks on QKD hardware.

3. **Endpoint attacks**
Attacks on Alice's or Bob's systems, key storage, software, or post-processing.

4. **Authentication or implementation weaknesses**
Weak authentication, poor randomness, or implementation flaws can be exploited.

Trusted key vs disrupted link

Confidentiality
If Eve causes too much disturbance, Alice and Bob detect it and reject the key.

Availability
Eve can disrupt the QKD session and deny service, but still fails to obtain a trusted secret.

Eve can cause disturbance without silently winning.

The quantum computer broke the maths. QKD forces the attacker to confront the physics, the implementation and the trust process.

QKD is not magic security. It is a change in security model and attack surface.

Experience Modes Map

How Phantom Q is used across structured learning, competition and experimentation

1 Missions

- Structured progression
- Learning journey
- Eventual storyline
- Solo or cooperative learning

2 Competition

- Player vs player
- Team vs team
- Red vs blue
- Trio vs trio challenge

3 Practice / Lab

- Isolated concept exploration
- Role training
- Hardware experimentation
- Low-pressure training

4 Classroom / Host

- Teacher-led use
- Role allocation
- Session monitoring
- Session control

Multiplayer is a capability across modes, not a separate mode by itself.

Phantom Q separates the way the platform is used without fragmenting the core security model.

Four Core Security Capabilities

The educational backbone that drives the Phantom Q platform

THE TECHNICAL BACKBONE OF PHANTOM Q

These four capabilities build in sequence and enable everything on the platform.

1 Foundations

- Text and image transformation
- Basic information protection

2 Symmetric

- Shared-secret security
- Attack on key distribution
- Real text/image protection

3 Asymmetric

- Public/private-key model
- Public information vs private information
- Mathematical attack challenge

4 QKD

- Trusted quantum key establishment
- Simulation and physical hardware
- Detectable interference

Every major gameplay, room and endgame decision must reinforce these four security capabilities.

Skill, Clearance and Access

Progression is earned through capability and trust, not decorative upgrades

1 Player Growth

- Knowledge
- Performance
- Consistency

3 Equipment

- Unlocks capability
- Does not replace expertise

Skill
↓
Demonstrated Capability
↓
Security Clearance
↓
Access to Restricted Facilities
↓
Advanced Systems and Responsibilities

2 Access Control

- Deeper headquarters areas
- More sensitive systems

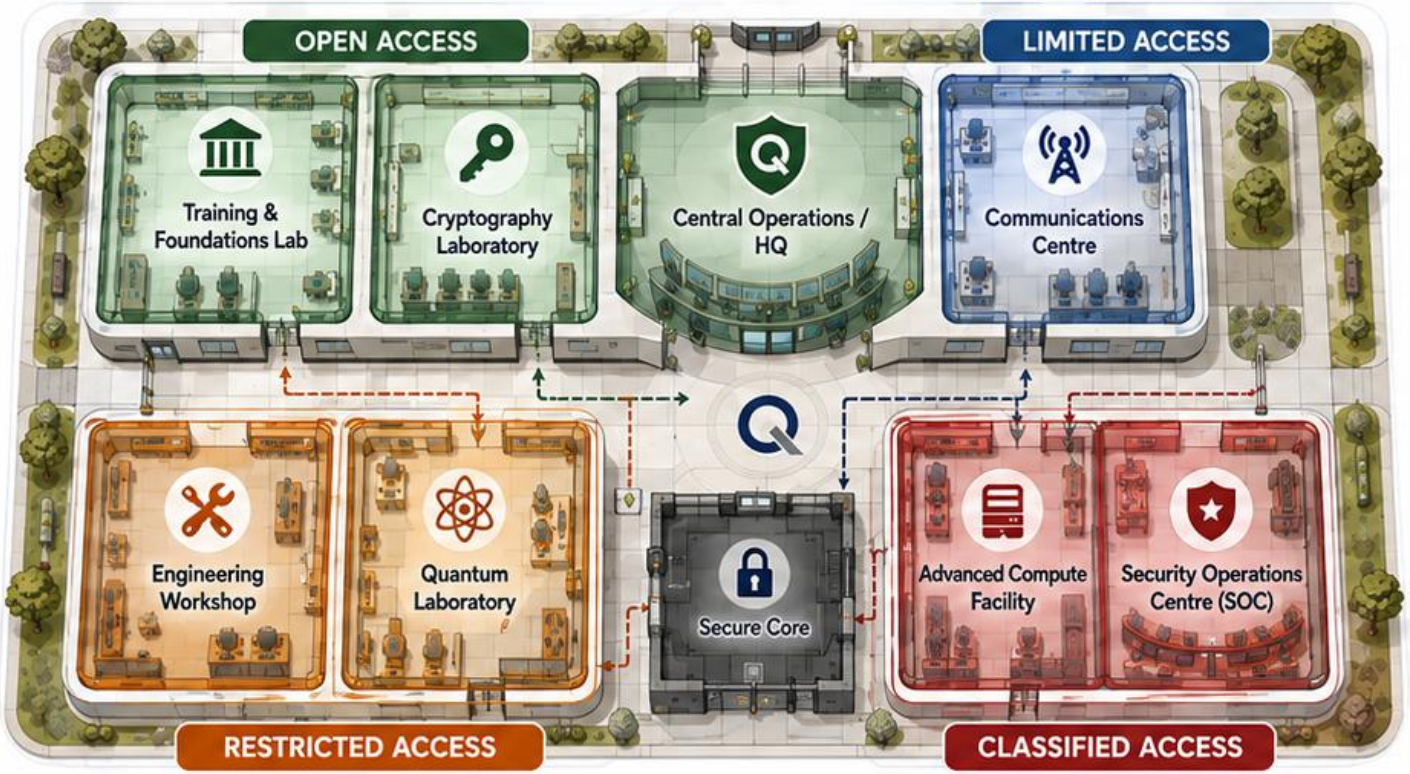
4 Future Consideration

- Advanced users may prove outside knowledge

The player advances by demonstrating cybersecurity capability, not by simply receiving stronger equipment.

Headquarters Access Progression

The same facility reveals deeper capability as trust and responsibility increase



1. **Initial Access**
   Training; Central HQ; Communications.
2. **Intermediate Access**
   Cryptography; SOC.
3. **Advanced Access**
   Red Team; Engineering; Quantum Lab.
4. **Restricted Endgame Access**
   Advanced Compute; Secure Core.

Progression is expressed through access to more critical Phantom Q functions inside the same headquarters.

QKD Threat-Surface Shift

QKD does not make security trivial; it changes what the attacker must target

1 Asymmetric stage

Mathematical hardness → Quantum-computer threat → Computational attack

The quantum computer threatens the mathematical challenge.

2 QKD stage

Quantum key establishment → Computational attack no longer targets the same type of secret

QKD shifts the problem rather than making security magic.

Core lesson

- QKD is not magic
- It changes the attack problem

Release 1 focus

- Intercept-resend
- Disturbance
- Key acceptance / rejection

Quantum channel

Intercept, disturb, or block photons.

Devices

Manipulate, replace, or exploit hardware.

Endpoints

Compromise hosts and key use.

Authentication

Impersonate or bypass identities.

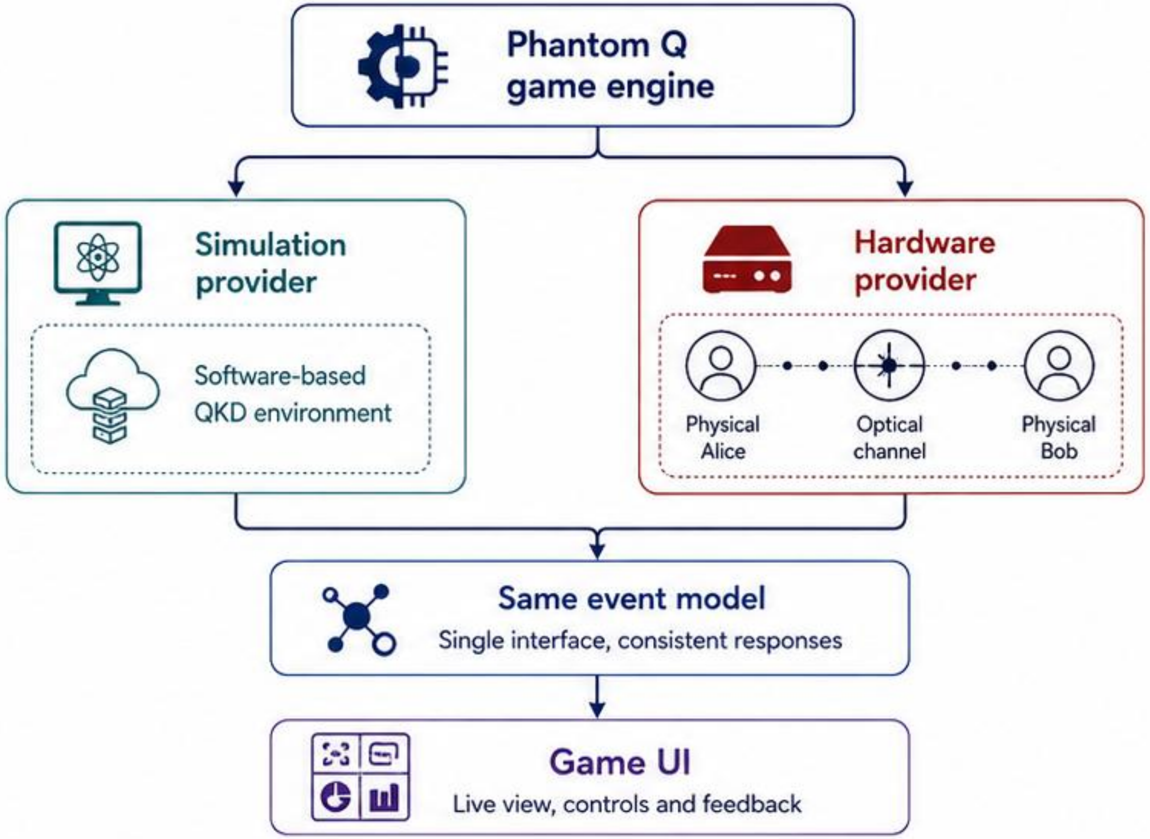
Availability

Disrupt, degrade, or deny service.

QKD shifts the attacker away from silent computational key recovery toward detectable interference and implementation-level threats.

Digital and Physical QKD Integration

The software platform and the real demonstrator form one operational QKD capability



1 Simulation

- Software-based QKD experience

2 Hardware

- Physical Alice
- Optical channel
- Physical Bob

3 Shared event model

- The same interface responds regardless of source

4 Operational advantage

- Simulation fallback when hardware is unavailable

Physical QKD hardware is not separate from Phantom Q; it is another live source of the same QKD system events.

Final Boss: The Phantom

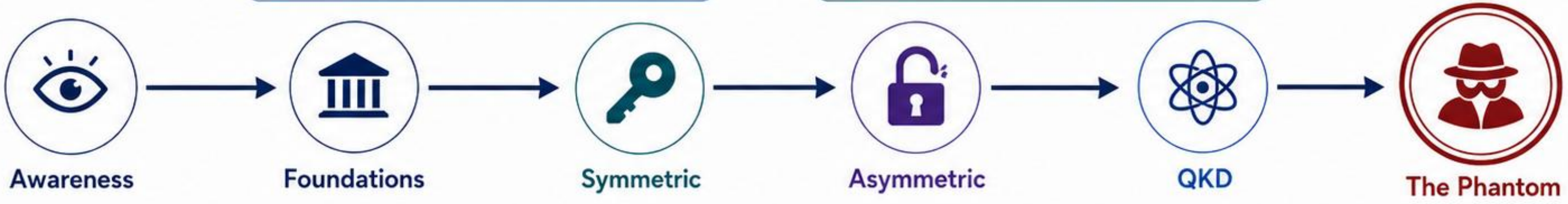
The endgame challenge that unifies the full Phantom Q learning journey

1 Why the Phantom

- Not simply Eve
- Broader adversarial challenge

2 Defender perspective

- Contain the current incident



3 Attacker perspective

- Breach the hardened security system

4 Design rule

- Final challenge must use established knowledge
- No disconnected boss mechanic

The final boss tests the learner's complete security understanding rather than introducing a new unrelated problem.

Persistent Adversary Model

Each campaign can end decisively while the broader threat environment continues to evolve

1. Incident defeated →
2. Learner grows →
3. New incident →
4. Stronger Phantom pattern →
5. Higher-level capability required

Campaign closure

- Current threat contained

Long-term progression

- New incidents
- New combinations of risk

Growth model

- Evolving defender and attacker skill

Why it matters

- Victory is real
- Learning continues

Players defeat the current Phantom incident, not cybersecurity itself.

Release Planning Structure

How the concept is divided into complete first-release scope and future expansion

1 Core Release

- Four core security concepts
- Core headquarters
- Flexible participation
- Basic missions/lab/competition structure
- QKD simulation and hardware connection

2 Limited Release

- Certain rooms in reduced scope
- Early social deduction
- Limited team formats

3 Next Release

- Deeper incident systems
- More advanced attacker options
- Expanded role-team behaviour

4 Future

- Advanced prior-knowledge testing
- Richer analytics
- Expanded HQ areas
- Larger campaigns

Scope Principle
Features are either complete within scope or deliberately deferred.

Features are either complete within scope or deliberately deferred; nothing should appear finished when it is not.

Full Cybersecurity Capability Universe

The broader Phantom Q scope beyond the initial cryptography and QKD pathway

PHANTOM Q CYBERSECURITY CAPABILITY UNIVERSE

11 **Professional Practice, Risk & Reporting**
ethics, risk, standards, communication

12 **Scripting & Automation**
Python, Bash, PowerShell-style automation for security tasks

1 **Cyber Foundations & Awareness**
CIA triad, threats, vulnerabilities, social engineering

10 **Quantum Security**
quantum threat, PQC, QKD, quantum-safe architecture

2 **Networking & Communications**
protocols, traffic, routing, secure transfer

9 **Cloud, Mobile, IoT, Hardware & OT**
devices, embedded systems, industrial and cloud security

3 **Systems, Identity & Endpoints**
accounts, permissions, MFA, hardening, endpoint security

8 **Incident Response & Forensics**
containment, recovery, evidence, investigation

4 **Cryptography**
symmetric, asymmetric, hashing, signatures, key management

7 **Defensive Security / SOC**
monitoring, logs, alerts, triage, threat intelligence

5 **Offensive Security**
reconnaissance, enumeration, exploitation, reporting

6 **Web, API & Software Security**
OWASP risks, sessions, access control, secure coding

1 Why expand the scope?

- Phantom Q can grow beyond a single encryption game
- The platform can become a hands-on cyber range and engineering learning environment

2 What still stays central?

- Release 1 remains anchored by Foundations, Symmetric, Asymmetric and QKD
- The wider scope protects long-term product direction

3 How external training informed this view

- MWR CyberSec contributed realistic attack–defence and red/blue/purple-team thinking
- Cisco contributed networking, endpoint, threat-management and defensive-skill breadth
- Kali / OffSec contributed offensive workflow, reconnaissance and tool-supported ethical hacking

4 Planning rule

- The full universe is captured now
- Release priorities are narrowed later without losing strategic direction

Takeaway: Phantom Q can evolve into a broad cyber range and engineering platform while keeping cryptography and quantum security as its distinctive starting pathway.

Headquarters Capability Expansion

Mapping the wider cybersecurity and quantum curriculum into Phantom Q Headquarters

1 Learn
- Training & Foundations Lab
- Cryptography Lab

2 Operate
- Communications Centre
- Central HQ

3 Attack / Test
- Red-Team Lab
- Advanced Compute Facility

4 Monitor / Respond
- SOC
- Secure Core for high-severity incidents

5 Engineer
- Engineering Workshop
- Quantum Wing

External training influences: Cisco informed communications and defensive operations; Kali / OffSec informed Red-Team capability; MWR CyberSec informed the integrated attack–defence structure.

Takeaway: Each Headquarters area represents a real cybersecurity or quantum-security capability, ensuring the map supports long-term learning growth rather than decorative navigation.

Ten Major Cybersecurity Disciplines

The full Phantom Q roadmap can be organised into ten major capability families

1. **Cyber Fundamentals & Awareness** — risk, threats, vulnerabilities, social engineering
2. **Networking & Communications** — protocols, traffic, routing, secure transfer
3. **Systems, Identity & Endpoints** — accounts, permissions, MFA, hardening, endpoint protection
4. **Cryptography** — symmetric, asymmetric, signatures, certificates, key management
5. **Offensive Security / Ethical Hacking** — reconnaissance, testing, exploitation, reporting
6. **Application, API & Software Security** — OWASP risks, secure coding, access control
7. **Defensive Security / SOC / Incident Response** — monitoring, detection, triage, containment and recovery
8. **Cloud, Mobile, IoT, Hardware & OT** — devices, embedded systems, cloud and industrial security
9. **Quantum Security** — quantum threat, PQC, QKD, quantum-safe architecture
10. **Professional Practice, Risk & Reporting** — ethics, standards, communication and decision-making

1 Why this grouping matters

- A broad vision prevents the product from becoming only an encryption demonstration
- The groups provide a stable structure for future Phantom Q expansion

2 What influenced the grouping

- Cisco influenced the breadth of networking, endpoint, threat-management and defensive domains
- MWR CyberSec influenced the attack–defence integration and realistic operational scope
- Kali / OffSec influenced the offensive-security and practitioner-methodology areas

3 Release 1 position

- Not all ten families need full implementation immediately
- They define the full planning horizon while the first release stays focused

Takeaway: The future Phantom Q roadmap can be planned as ten disciplined capability families, allowing the platform to grow methodically without losing focus.

Attack, Defence and Purple-Team Model

How Phantom Q can train offensive, defensive and collaborative cybersecurity thinking

RED TEAM

- reconnaissance
- enumeration
- vulnerability testing
- exploitation
- adversary simulation
- reporting

BLUE TEAM

- monitoring
- log analysis
- detection
- alert triage
- containment
- recovery

PURPLE TEAM

- shared validation
- attack replay
- detection improvement
- coverage testing
- lessons learned
- continuous improvement

Authorised attack vs malicious activity

Authorised Eve / Red-Team activity

- approved scope
- ethical testing
- learning and improvement

Phantom / Imposter activity

- deceptive or unauthorised
- real compromise
- security incident

1 **What MWR contributed**

- A realistic attack-defence mindset
- Red, blue and purple-team thinking

2 **What Cisco contributed**

- Network defence, cyber threat management and incident response breadth

3 **What Kali / OffSec contributed**

- Tool-backed ethical hacking workflow and practitioner methodology

4 **Phantom Q Implication**

- Attack capability is not automatically malicious
- Authorisation determines whether a capability is used for learning or compromise

Takeaway: Phantom Q should teach attack, defence and purple-team collaboration as connected cybersecurity practices, while clearly separating authorised testing from real adversarial compromise.

Offensive Security Capability

Positioning Kali, OffSec and adversarial workflow inside the future Phantom Q attack laboratory

1 **Reconnaissance**
OSINT, discovery, target awareness

2 **Enumeration**
services, users, paths, exposed surfaces

3 **Vulnerability Testing**
validation, weakness identification

4 **Exploitation Practice**
controlled exploitation and access testing

5 **Post-Validation**
impact verification, privilege checks, evidence capture

6 **Reporting & Ethics**
scope, findings, recommendations, authorised conduct

Tool-supported practice, not tool dependency

- Kali Linux is a practitioner environment
- OffSec contributes offensive methodology
- Phantom Q should teach capability, not memorisation of tools

1 What Kali / OffSec contributed

- offensive workflow
- practitioner methodology
- tool-backed ethical hacking

2 What MWR reinforced

- realistic adversarial testing
- attack–defence mindset
- reporting and improvement

3 Planning principle

- reconnaissance, testing and exploitation are features
- specific tools support those features
- the platform should stay methodology-led

4 Phantom Q implication

- Eve hacking sits inside an authorised learning structure
- offensive skill must remain bounded by scope and evidence
- attack knowledge strengthens defence

Takeaway: Phantom Q should teach offensive security as a structured, authorised capability built on methodology, evidence and ethical practice rather than isolated tools.

Defensive Security and SOC Capability

How monitoring, evidence and incident response expand the Phantom Q defensive learning model

1. **Monitoring** — visibility across systems and activity
2. **Log Analysis** — patterns, baselines and evidence review
3. **Alert Triage** — prioritisation, severity and validation
4. **Threat Intelligence** — indicators, context and adversary understanding
5. **Investigation** — timeline building, root cause and scope
6. **Containment & Recovery** — response actions, restoration and resilience
7. **Reporting & Lessons Learned** — communication, improvement and defensive maturity

SOC outcomes

- detect suspicious behaviour
- convert events into evidence
- guide containment and recovery

1 What Cisco contributed

- networking breadth
- endpoint and threat-management awareness
- defensive and incident-response structure

2 What MWR reinforced

- attacker behaviour informs detection
- realistic operations mindset
- purple-team improvement loop

3 Where this fits in Headquarters

- SOC monitors
- Secure Core handles high-severity systems
- Communications Centre supports network context

4 Phantom Q implication

- defence is not passive
- learners must observe, analyse and respond
- evidence-based decisions become part of gameplay

Takeaway: The Phantom Q defensive path should turn monitoring, evidence and incident response into active learner capabilities rather than background system behaviour.

Quantum Security Wing

Mapping the wider quantum-security scope beyond the initial QKD demonstrator

PHANTOM Q QUANTUM SECURITY WING

1 Quantum Foundations
- qubits
- measurement
- superposition
- no-cloning

2 Quantum Computing
- circuits
- gates
- algorithms
- noise

3 Quantum-Safe Cybersecurity
- PQC
- crypto inventory
- migration
- long-term risk

4 Quantum Communications / QKD
- BB84
- sifting
- QBER
- trusted key establishment

5 Quantum Engineering
- optics
- detectors
- control electronics
- calibration

6 Quantum Security Operations
- attacks
- monitoring
- device status
- response

Release 1 anchor
- Foundations, Symmetric, Asymmetric and QKD remain the starting pathway
- the wider quantum wing protects long-term direction

Takeaway: The Quantum Security Wing lets Phantom Q preserve QKD as its starting differentiator while creating a coherent structure for broader quantum capability growth.

1 Why this wing matters
- quantum is larger than encryption;
- QKD is a major capability, not the whole vision

2 How it links to Headquarters
- Quantum Wing connects to Engineering Workshop;
- Advanced Compute supports quantum-computing ideas;
- Secure Core uses high-value quantum-safe systems

3 Planning principle
- capture the full wing now;
- narrow the build later;
- keep the architecture open for growth

4 Phantom Q implication
- the platform can grow from a QKD pathway into a broader quantum-security and quantum-engineering environment

Quantum Computing Capability

Defining the computing side of the quantum pathway without reducing it to a universal hacking machine

A Classical vs Quantum

CLASSICAL	QUANTUM
Bits (0 or 1)	Qubits (superposition)
Deterministic logic	Probabilistic measurement
Conventional computation (step-by-step)	Quantum speed-up for specific tasks

B Quantum computing learning areas

1. **Qubits & States** — Understand qubits, superposition, entanglement and quantum states.
2. **Gates & Circuits** — Quantum gates, circuits, and building algorithmic structures.
3. **Measurement & Interference** — Measurement, collapse, interference patterns and probability outcomes.
4. **Algorithms — Shor, Grover, quantum search concepts** — Key algorithms and how they provide quantum advantage.
5. **Noise, Error Correction & Fault Tolerance** — Noise sources, error correction techniques and fault-tolerant architectures.
6. **Security Relevance — why some cryptographic assumptions are threatened** — Impact on cryptographic assumptions and security models.

1 Why this matters to security

- quantum computing changes specific hard problems
- security learners must understand the threat model behind it

2 What it does not mean

- a quantum computer is not a universal hacking machine
- different algorithms solve different classes of problems

3 How it links to Phantom Q

- Advanced Compute Facility supports this pathway
- Asymmetric security leads into the quantum-computer threat
- QKD and PQC become different responses

4 Planning principle

- teach concepts first
- attach deeper mathematics later
- keep the capability understandable for mixed audiences

Takeaway: Phantom Q should teach quantum computing as a real capability area that explains the security threat landscape, rather than presenting it as a vague all-powerful attacker tool.

Quantum-Safe Cybersecurity

Positioning PQC and QKD as complementary responses to the quantum-era security challenge

Quantum Threat

Advances in quantum computing threaten current public-key cryptography and digital security

PQC

- classical algorithms
- software migration
- protects digital systems
- replaces vulnerable public-key assumptions

QKD

- quantum communication
- hardware / network integration
- trusted key establishment
- supports secure applications

Quantum-Safe Architecture

- PQC + QKD can coexist
- different problems, different deployment models
- hybrid approaches are possible

Takeaway: Phantom Q should teach quantum-safe cybersecurity as a broader architecture that includes both post-quantum cryptography and QKD.

1 Why this matters

- quantum-safe security is larger than QKD alone
- learners should understand multiple defensive paths

2 What NIST / CISA emphasised

- start migration planning now
- build cryptographic inventories
- prepare for long-term transition

3 What ETSI clarified

- QKD sits inside a broader quantum-safe ecosystem
- interfaces and hybrid models matter

4 Planning principle

- do not teach QKD as the only answer
- show how PQC and QKD solve different security problems

Cryptographic Migration and Quantum Readiness

How organisations identify vulnerable cryptography and plan a quantum-safe transition

1 Crypto Inventory
- find certificates, VPNs, TLS, signatures, key exchanges

2 Data & Longevity
- classify sensitive information
- identify long-term confidentiality needs

3 Quantum Exposure
- locate RSA / ECC dependencies
- map vulnerable systems

4 Prioritise Migration
- locitcal systems first
- plan replacement psthways

5 Transition & Validate
- deploy PQC / quantum-safe updates
- test interoperability and trust

6 Ongoing Readiness
- monitor progress
- track new risks and remaining gaps

Harvest-now, decrypt-later
- captured data may be decrypted in the future
- long-life secrets need early protection

Takeaway: Quantum readiness begins with discovering where cryptography is used, understanding long-term risk, and planning controlled migration.

1 What NIST / CISA emphasised
- build cryptographic inventories
- identify dependencies and risks
- plan and execute migration strategies

2 Why this matters to Headquarters
- Secure Core protects high-value systems
- SOC needs visibility into crypto risk
- Communications Centre tracks operational impact

3 Planning principle
- quantum readiness starts before deployment
- discovery comes before replacement

4 Phantom Q implication
- learners should assess before they upgrade
- migration becomes a strategic capability

QKD System Architrecture

Showing how QKD produces trusted key material before it reaches a secure application

1. **Quantum Channel** → Alice sends quantum states; Bob receives over an insecure quantum link.
2. **Detection Events** → raw measurement outcomes at Bob.
3. **Sifting** → Alice and Bob compare bases (public channel) and keep matching bits.
4. **QBER / Disturbance Check** → estimate error rate (QBER); verify channel is within acceptable limits.
5. **Reconciliation** → correct errors between Alice and Bob to align their bit strings.
6. **Privacy Amplification** → compress the key to remove any information an eavesdropper may have.
7. **Key Management** → store, protect, rotate and distribute keys via a key management system.
8. **Secure Application** → trusted keys are used by applications for encryption and secure services.

Scene 5 anchor

- manual basis choice
- sifted key formation
- verification workflow

Takeaway: QKD should be taught as a full trusted-key system that flows from quantum exchange through key processing into secure application use.

1 What ETSI clarified

- QKD interfaces with applications through key management
- QKD is a system, not only a photon demonstration

2 What this is not

- QKD does not directly send the file
- it does not remove the need for application security

3 Why this matters

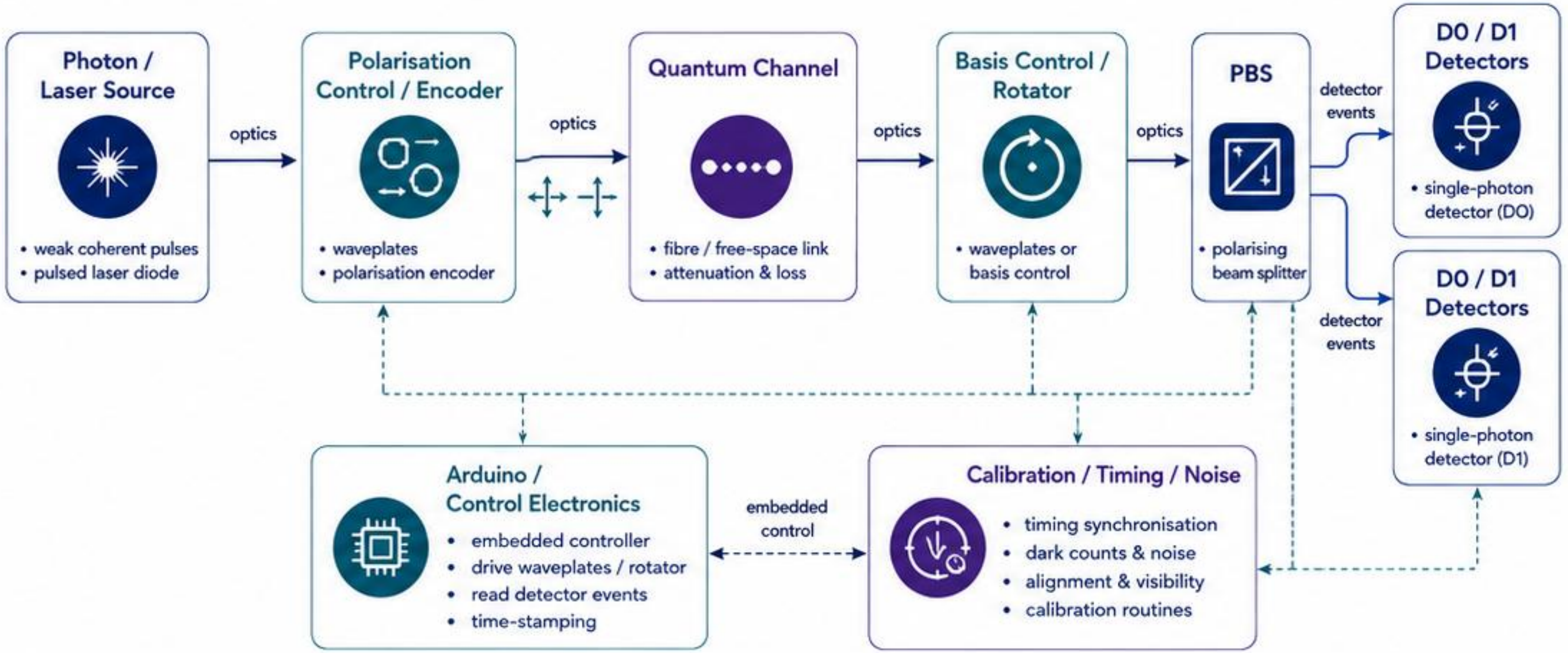
- it separates protocol stages
- it shows where trust decisions happen

4 Planning principle

- teach the complete key lifecycle
- connect generated keys to practical secure use

Physical QKD Engineering

Connecting the software pathway to the real demonstrator hardware and optical bench



Demonstrator value

- turns theory into hardware
- supports calibration and troubleshooting
- links cyber learning with engineering practice

Takeaway: The physical QKD demonstrator makes Phantom Q a quantum-engineering platform by linking optics, electronics, calibration and secure-key theory.

1 What the physical build contributes

- real optics
- real detectors
- real control electronics

2 Why this matters

- protocol security depends on implementation quality
- hardware faults and security behaviour can look similar

3 How it links to Headquarters

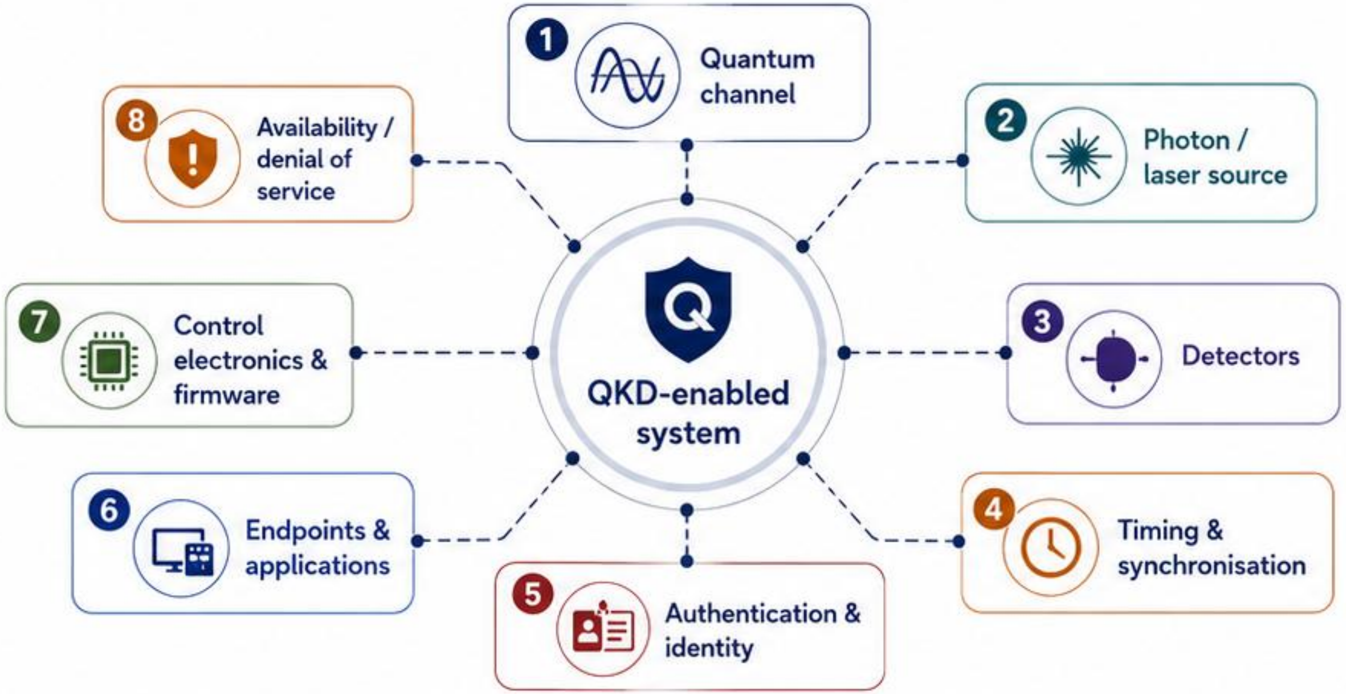
- Engineering Workshop supports embedded systems
- Quantum Wing supports QKD experimentation
- SOC can later interpret operational anomalies

4 Planning principle

- teach the system as engineering, not only theory
- keep room for calibration, measurement and fault analysis

Quantum Threat Surfaces

Showing where security and engineering risks move once QKD becomes part of a real system



THREAT-SURFACE CATEGORIES

PROTOCOL | DEVICES | OPERATIONS | APPLICATIONS

1 Why this matters

- QKD changes the attack surface
- the protocol can be sound while implementation risk remains
- cyber and engineering security both matter

2 Security lesson

- errors are not automatically proof of Eve
- some anomalies come from hardware faults or noise
- learners must distinguish attack signals from system imperfections

3 Phantom Q implication

- teach the whole system, not only BB84 theory
- link SOC awareness with engineering troubleshooting
- show where attackers shift their attention

4 Planning principle

- treat quantum security as architecture
- protect devices, identity, timing and operations
- use QKD inside a wider defensive design

Takeaway: QKD does not remove security work; it redistributes risk across the channel, devices, operations and applications.

Quantum Red, Blue and Purple Teams

Framing advanced quantum-security training as coordinated attack, defence and improvement activity

Quantum Red Team

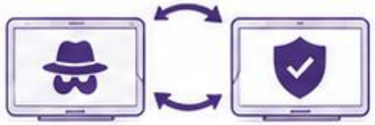
- probe QKD implementations
- test source, detector and endpoint weaknesses
- challenge authentication and trust
- attempt disruption, interception or implementation abuse
- document offensive findings

Quantum Blue Team

- monitor QBER and detector behaviour
- verify device status and link trust
- identify anomalies and respond
- protect keys, endpoints and operations
- maintain trusted service

Quantum Purple Team

- coordinate attack and defence learning
- run controlled tests against monitoring
- improve detection rules and procedures
- translate findings into better engineering and operations
- close security gaps



test -> detect -> improve

1 Why this is useful

- turns QKD into a real security training environment
- connects cyber operations with quantum engineering
- supports advanced team-based gameplay

2 Training alignment

- Red / Blue / Purple thinking is common in modern cybersecurity practice
- offensive and defensive capability should reinforce each other
- Phantom Q can adapt this model to quantum security

3 Phantom Q implication

- future missions can pit quantum attackers against defenders
- teachers can run controlled challenge exercises
- the platform grows beyond a static physics demonstration

Operational focus | implementation security | monitoring | incident response | continuous improvement

Takeaway: Quantum security can be taught as a living attack-defence discipline, not only as a protocol explanation.

Long-Term Phantom Q Roadmap

Showing how the first cryptography release can expand into a broader cyber and quantum security platform

1 Release 1 — Core Foundation

- Foundations, symmetric, asymmetric and QKD pathway
- Alice / Bob / Eve roles
- headquarters concept and guided learning
- physical QKD demonstrator link

2 Next Release — Expanded Security Operations

- SOC activity and incident response
- awareness / imposter scenarios
- refined attack-defence competition
- improved room-based headquarters interaction

3 Growth Phase — Broader Cybersecurity

- networking and communications security
- web / API and application security
- offensive labs with Kali-style tool support
- engineering workshop, IoT and hardware security
- reporting and evidence workflows

4 Advanced Future — Quantum-Safe Ecosystem

- post-quantum cryptography and migration planning
- quantum red / blue / purple team exercises
- QKD operations, key management and architecture
- quantum networking and advanced compute concepts
- persistent Phantom incidents across the platform

1 Scope discipline

- build complete features, not unfinished placeholders
- separate core release from future vision

2 External capability influences

- MWR informed attack-defence and offensive thinking
- Cisco informed networking, SOC and defensive pathways
- Kali / OffSec informed offensive methodology and tool-supported labs

3 Identity principle

- Phantom Q remains cyber plus quantum
- QKD and engineering stay as key differentiators

4 Planning principle

- the roadmap prevents drift
- each release should add coherent capability families

ROADMAP DIRECTION | LEARN | OPERATE | ATTACK | DEFEND | ENGINEER

Takeaway: Phantom Q should grow from a focused cryptography release into a wider cyber and quantum-security platform without losing its identity or engineering depth.